Saddle-Dominated Capacity of the LSE Dense Associative Memory

Abstract

[Placeholder.] The log-sum-exp Dense Associative Memory [3] stores M unit vectors $\boldsymbol{\xi}^\mu$ on the N-sphere with kernel sharpness β_{net}. Ramsauer et al. proved $\alpha_c = 1/2$ at $T = 0$ by modelling the inter-pattern overlaps as Gaussian. Replacing the Gaussian by the exact spherical density opens an interior saddle in the partition sum at $\varphi^*(\beta_{\text{net}})$, with value $g_{\max}(\beta_{\text{net}})$. The boundary becomes $\alpha_c(\beta_{\text{net}}, T) = \beta_{\text{net}}[1 - f_{\text{ret}}(T)] - g_{\max}(\beta_{\text{net}})$. At $\beta_{\text{net}} = 1$ — the value we use in the Monte Carlo — the saddle is the golden ratio $\varphi^* = (\sqrt{5} - 1)/2$ and $g_{\max} \approx 0.3774$, giving $\alpha_c(0) \approx 0.6226$, not 0.5. Ramsauer's 1/2 is β_{net}-independent (an extreme-value cancellation), the exact-density boundary is not. We confirm the corrected value with three Monte Carlo variants (direct, truncated, hybrid) at N up to 100.

1 Introduction

A Dense Associative Memory [1] stores M patterns $\boldsymbol{\xi}^\mu \in \mathbb{R}^N$, $\mu = 1, \ldots, M$, as minima of an energy function $H(\boldsymbol{x})$. Retrieval of $\boldsymbol{\xi}^1$ from a noisy cue is the Metropolis dynamics of $\boldsymbol{x}$ under H at external temperature T. Both the patterns and the chain live on the N-sphere of radius $\sqrt{N}$.

The log-sum-exp (LSE) variant of Ramsauer et al. uses

$$H(\boldsymbol{x}) = -\beta_{\text{net}}^{-1} \ln \sum_{\mu=1}^{M} \exp(\beta_{\text{net}}\, \boldsymbol{x} \cdot \boldsymbol{\xi}^\mu), \qquad \beta_{\text{net}} = 1 \text{ throughout}, \tag{1}$$

and stores $M = \mathrm{e}^{\alpha N}$ patterns at load $\alpha = \ln M/N$. The question is: what is the largest α at which the chain initialised at $\boldsymbol{\xi}^1$ stays close to $\boldsymbol{\xi}^1$?

Ramsauer et al. answered $\alpha_c = 1/2$ at $T = 0$. The proof modelled the inter-pattern overlap $\varphi_{1\mu} = \boldsymbol{\xi}^1 \cdot \boldsymbol{\xi}^\mu / N$ as a Gaussian with variance $1/N$. The Gaussian is the leading term of the exact spherical density

$$f(\varphi) \;\propto\; (1 - \varphi^2)^{(N-3)/2}, \qquad \varphi \in [-1, 1], \tag{2}$$

near $\varphi = 0$, but the two densities differ near $\varphi = 1$. That tail matters: it sets the typical maximum of M samples and the saddle of the partition sum.

This paper does three things. First, we re-derive the boundary using the exact density (2). An interior saddle of the integrand appears at a point $\varphi^*(\beta_{\text{net}})$ that depends on the kernel sharpness; at $\beta_{\text{net}} = 1$ — the value we use throughout — it is the golden ratio $\varphi^* = (\sqrt{5} - 1)/2$, with saddle value $g_{\max} \approx 0.3774$, and the boundary shifts to $\alpha_c(0) = 1 - g_{\max} \approx 0.6226$. Second, we describe three Monte Carlo schemes (direct, truncated, hybrid) and explain why a naive "smart" variant that refreshes its pattern cache fails near α_c. Third, we confirm the corrected boundary numerically at $N = 100$ by an escape-time measurement and by a ϕ residual heatmap.

2 Theory

Free energy split. For a chain in the basin of $\boldsymbol{\xi}^1$, the LSE sum is dominated by the $\mu = 1$ term, $H(\boldsymbol{x})/N \approx -\varphi_1$ with $\varphi_1 = \boldsymbol{\xi}^1 \cdot \boldsymbol{x}/N$. Define the energy per spin relative to the retrieval baseline, $u(\varphi_1) = 1 - \varphi_1$, and the $(N-1)$-sphere entropy per spin $s(\varphi_1) = \frac{1}{2}\ln(1-\varphi_1^2)$. The retrieval free energy is the minimum of $f(\varphi_1, T) = u - Ts$:

$$f_{\rm ret}(T) = (1-\phi_{\rm eq}) - \tfrac{T}{2}\ln(1-\phi_{\rm eq}^2), \qquad \phi_{\rm eq}(T) = \tfrac{1}{2}(-T + \sqrt{T^2+4}). \tag{3}$$

For a chain uncorrelated with $\boldsymbol{\xi}^1$ ($\varphi_1 \approx 0$) the sum in (1) is dominated by the $M-1$ spurious terms $S = \sum_{\mu\neq 1}\exp(\beta_{\rm net} N\varphi_{1\mu})$. Replacing the sum by an integral with density (2) and using $M = e^{\alpha N}$:

$$S \;\approx\; e^{\alpha N}\int_{-1}^{1}\exp\!\Big[N\left(\tfrac{1}{2}\ln(1-\varphi^2) + \beta_{\rm net}\,\varphi\right)\Big]\mathrm{d}\varphi \;\equiv\; e^{\alpha N}\int e^{Ng(\varphi;\beta_{\rm net})}\mathrm{d}\varphi. \tag{4}$$

$g(\varphi;\beta_{\rm net}) = \frac{1}{2}\ln(1-\varphi^2) + \beta_{\rm net}\,\varphi$ has an interior saddle at $g'(\varphi^*) = 0$, i.e. $\beta_{\rm net}(1-\varphi^{*\,2}) = \varphi^*$, with positive root

$$\varphi^*(\beta_{\rm net}) \;=\; \frac{-1+\sqrt{1+4\beta_{\rm net}^2}}{2\beta_{\rm net}}. \tag{5}$$

The saddle value is

$$g_{\max}(\beta_{\rm net}) \;=\; g(\varphi^*;\beta_{\rm net}) \;=\; \tfrac{1}{2}\ln(1-\varphi^{*\,2}) + \beta_{\rm net}\,\varphi^*. \tag{6}$$

Hence $N^{-1}\ln S \to \alpha + g_{\max}(\beta_{\rm net})$, the noise-floor energy per spin is $u_{\rm noise} = 1 - [\alpha + g_{\max}(\beta_{\rm net})]/\beta_{\rm net}$, and the boundary at the crossover $u_{\rm noise} = f_{\rm ret}(T)$ is

$$\boxed{\alpha_c(\beta_{\rm net}, T) \;=\; \beta_{\rm net}\,[1 - f_{\rm ret}(T)] - g_{\max}(\beta_{\rm net}).} \tag{7}$$

Gaussian and support-edge variants. Two well-known prior formulas come from substituting different objects into (4) and using its *support edge* instead of the interior saddle.

Gaussian density. Replace $f(\varphi)$ by the moment-matched Gaussian $\sqrt{N/2\pi}\,\exp(-N\varphi^2/2)$. The integrand exponent becomes $g_G(\varphi;\beta_{\rm net}) = -\frac{1}{2}\varphi^2 + \beta_{\rm net}\varphi$, monotone on $[-1,1]$ for $\beta_{\rm net} \le 1$. The integral is dominated by the typical maximum of M Gaussian samples $\varphi_{\max}^{(G)}(\alpha) = \sqrt{2\alpha}$; at this edge $g_G(\varphi_{\max}^{(G)}) = \beta_{\rm net}\sqrt{2\alpha} - \alpha$, so $N^{-1}\ln S \to \beta_{\rm net}\sqrt{2\alpha}$ and $u_{\rm noise}^{(G)} = 1 - \sqrt{2\alpha}$. Solving $u_{\rm noise}^{(G)} = f_{\rm ret}(T)$:

$$\alpha_c^{\rm G}(T) \;=\; \tfrac{1}{2}\big[1 - f_{\rm ret}(T)\big]^2, \qquad \alpha_c^{\rm G}(0) = \tfrac{1}{2}. \tag{8}$$

$\beta_{\rm net}$ cancels — both retrieval and spurious-extreme energies are $N\!\cdot\!\beta_{\rm net}$ times an overlap, and the $\beta_{\rm net}^{-1}$ prefactor in H removes it. This is Ramsauer's threshold.

Exact density, support edge. Keep $f(\varphi)$ but use the support edge of the exact density $\varphi_{\max}(\alpha) = \sqrt{1-e^{-2\alpha}}$ (from $M\cdot \Pr_f(\varphi > \varphi_{\max}) = 1$). At this edge $\frac{1}{2}\ln(1-\varphi_{\max}^2) = -\alpha$, so $g(\varphi_{\max}) = -\alpha + \beta_{\rm net}\,\varphi_{\max}$, $N^{-1}\ln S \to \beta_{\rm net}\,\varphi_{\max}(\alpha)$ and $u_{\rm noise}^{(\rm ex)} = 1 - \varphi_{\max}(\alpha)$. Solving $1 - \varphi_{\max}(\alpha_c^{\rm bd}) = f_{\rm ret}(T)$:

$$\alpha_c^{\rm bd}(T) \;=\; -\tfrac{1}{2}\ln\!\big[1 - (1 - f_{\rm ret}(T))^2\big], \qquad \alpha_c^{\rm bd}(0) = +\infty. \tag{9}$$

This is the formula that works for the LSR kernel where g is monotone on the support [2]; for LSE it is correct only when the interior saddle φ^* lies outside the support $[0, \varphi_{\max}(\alpha)]$, i.e. for $\alpha < -\frac{1}{2}\ln(1-\varphi^{*\,2}) \approx 0.241$ at $\beta_{\rm net} = 1$. Above this threshold the interior saddle (7) takes over. Both (8) and (9) are $\beta_{\rm net}$-independent for the same reason: the support edge replaces the $\beta_{\rm net}$-dependent saddle.

Cusp picture. At fixed φ_1 the spurious sum evaluates to $\mathrm{e}^{N(\alpha+g_{\max}(\beta_{\mathrm{net}}))}$ to leading order, independent of φ_1. Inside the LSE log-sum, $\beta_{\mathrm{net}}^{-1}\ln[\mathrm{e}^{\beta_{\mathrm{net}}N\varphi_1}+S] \to \max\big(\varphi_1,\ [\alpha+g_{\max}(\beta_{\mathrm{net}})]/\beta_{\mathrm{net}}\big)$ at large N, so the leading-order free energy per spin reads

$$\frac{F(\varphi_1)}{N} \;=\; -\max\!\Big(\varphi_1,\ \frac{\alpha+g_{\max}(\beta_{\mathrm{net}})}{\beta_{\mathrm{net}}}\Big) \;-\; \tfrac{T}{2}\ln(1-\varphi_1^2), \tag{10}$$

with a kink ("cusp") at $\varphi_c(\alpha,\beta_{\mathrm{net}}) = [\alpha+g_{\max}(\beta_{\mathrm{net}})]/\beta_{\mathrm{net}}$. As T rises the basin minimum $\phi_{\mathrm{eq}}(T)$ slides toward the cusp; at $\phi_{\mathrm{eq}}(T)=\varphi_c$ the local minimum disappears, defining the single-pattern spinodal

$$\alpha_c^{\mathrm{sp}}(\beta_{\mathrm{net}},T) \;=\; \beta_{\mathrm{net}}\,\phi_{\mathrm{eq}}(T) - g_{\max}(\beta_{\mathrm{net}}), \qquad T_{\mathrm{sp}}(\alpha,\beta_{\mathrm{net}}) \;=\; \frac{\beta_{\mathrm{net}}^2 - (\alpha+g_{\max}(\beta_{\mathrm{net}}))^2}{\beta_{\mathrm{net}}\,(\alpha+g_{\max}(\beta_{\mathrm{net}}))}. \tag{11}$$

$\alpha_c^{\mathrm{sd}}(\beta_{\mathrm{net}},T)$ and $\alpha_c^{\mathrm{sp}}(\beta_{\mathrm{net}},T)$ meet at the endpoint $(\beta_{\mathrm{net}} - g_{\max}(\beta_{\mathrm{net}}),\,0)$.

Examples and the $\beta_{\mathrm{net}}\to\infty$ limit. Four cases of (5), (6), and (7) at $T=0$:

β_{net}	$\varphi^*(\beta_{\mathrm{net}})$	$g_{\max}(\beta_{\mathrm{net}})$	$\alpha_c(\beta_{\mathrm{net}},0)$	note
0.5	$\sqrt{2}-1\approx 0.414$	≈ 0.113	≈ 0.387	—
1	$(\sqrt{5}-1)/2\approx 0.618$ (golden ratio)	≈ 0.3774	≈ 0.6226	MC value
2	$(\sqrt{17}-1)/4\approx 0.781$	≈ 1.091	≈ 0.909	—
$\to\infty$	$\to 1-1/(2\beta_{\mathrm{net}})$	$\beta_{\mathrm{net}}-\frac{1}{2}(\ln\beta_{\mathrm{net}}+1)$	$\frac{1}{2}(\ln\beta_{\mathrm{net}}+1)$	log growth

Capacity grows logarithmically with β_{net}. The Ramsauer prediction $\alpha_c^{\mathrm{G}}=1/2$ is β_{net}-independent: its derivation replaces the integrand in (4) by its support edge $\sqrt{2\alpha}$, and the factor β_{net} then cancels between retrieval and spurious terms. The interior saddle restores the β_{net} dependence; the β_{net}-independence of the Gaussian-edge answer is itself the signature of the support-edge replacement. The MC numerics in Sections 3 and 4 fix $\beta_{\mathrm{net}}=1$, where $\alpha_c(1,0)\approx 0.6226$ is the value we verify. Figure 1 illustrates (10) at $\alpha=0.50$, $\beta_{\mathrm{net}}=1$.

3 Monte Carlo schemes

All chains run on the $\sqrt{N}$-sphere at $\beta_{\mathrm{net}}=1$ with Metropolis updates. The variants differ in how they handle the $M=\mathrm{e}^{\alpha N}$ patterns.

Direct. Store all M patterns explicitly and evaluate (1) over all of them at every step. Memory and compute cost $\sim MN$. Feasible up to $N\approx 25$ at $\alpha\approx 0.7$ ($M\approx 4\times 10^7$); impossible at $N=100$, $\alpha=0.6$ ($M\approx 10^{26}$).

Truncated. Generate only the patterns with overlap $\varphi_{1\mu}>\varphi_{\mathrm{keep}}$ to the target $\boldsymbol{\xi}^1$. The cutoff is chosen so the expected count above it equals a budget $K\sim 10^4$; the actual count is Poisson with mean $M\cdot\Pr(\varphi>\varphi_{\mathrm{keep}})$, and the K overlap values are drawn from the Beta tail. The patterns below φ_{keep} are not generated; their contribution enters the LSE log-sum as the analytic constant $\ln C_{\mathrm{bulk}} = \alpha N + \ln\int_{-1}^{\varphi_{\mathrm{keep}}} f(\varphi)\,\mathrm{e}^{N\varphi}\,\mathrm{d}\varphi$. The retained set is fixed once per disorder. Truncated MC at $N=25$ matches direct MC on the basin boundary across the full α range we measure, so the $\varphi<\varphi_{\mathrm{keep}}$ patterns are genuinely unnecessary for the basin physics, not silently dropped. We use truncated for the wide heatmap.

Hybrid. Same kept-tail + analytic-bulk decomposition as truncated, with two changes: the cutoff is fixed *at the cusp position* $\varphi_c(\alpha,\beta_{\mathrm{net}}) = [\alpha+g_{\max}(\beta_{\mathrm{net}})]/\beta_{\mathrm{net}}$ from (10) rather than chosen by budget, and the bulk integral is evaluated by its saddle (6) instead of numerically. Patterns with $\varphi_{1\mu}>\varphi_c$ are kept live; the contribution of those with $\varphi_{1\mu}<\varphi_c$ enters the LSE

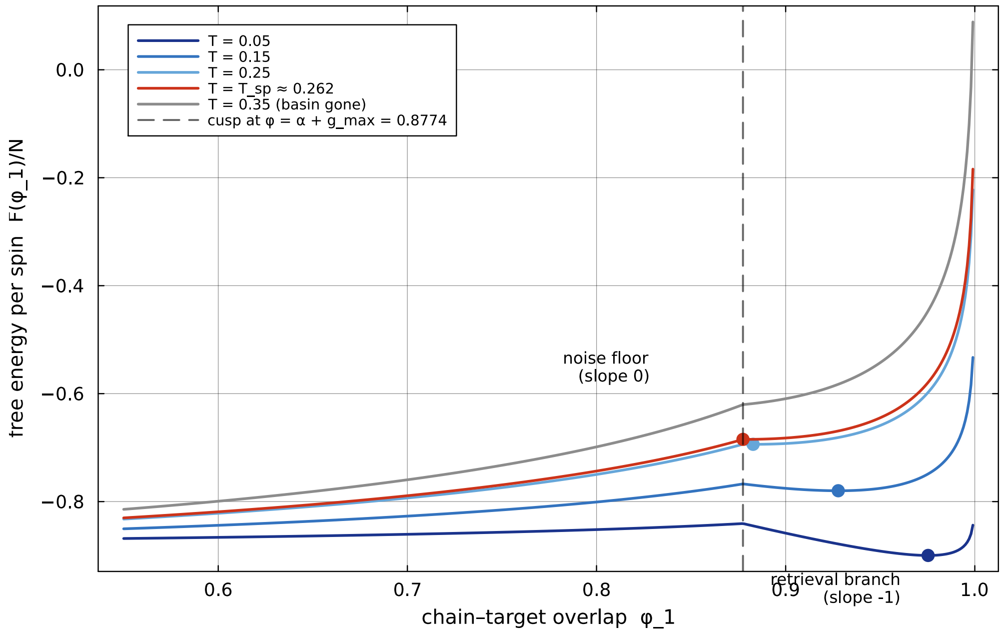

Figure 1: Leading-order free energy per spin $F(\varphi_1)/N$ from (10) at $\alpha = 0.50$, $\beta_{\text{net}} = 1$, for several T. The cusp at $\varphi_c = \alpha + g_{\max} \approx 0.877$ is the boundary between the retrieval branch (slope -1) and the noise-floor branch (slope 0). Dots: basin minima $\phi_{\text{eq}}(T)$. At $T = T_{\text{sp}}(0.50, 1) \approx 0.262$ the minimum reaches the cusp; above this T the basin no longer exists.

log-sum as the $\boldsymbol{x}$-independent number $M\,\mathrm{e}^{Ng_{\max}(\beta_{\text{net}})}$ (state-independent by rotational symmetry of random patterns, after averaging over their orthogonal directions to $\boldsymbol{\xi}^1$).

The live count K_{cusp} is drawn from a Poisson with mean $M \cdot \Pr_f(\varphi > \varphi_c)$. Live overlap values are sampled from the spherical density (2) restricted to $\varphi > \varphi_c$. In the rescaled variable $y = (1+\varphi)/2$ on $[0,1]$, (2) becomes Beta$((N-1)/2,\,(N-1)/2)$ — the change of variable $(1-\varphi^2)^{(N-3)/2}\mathrm{d}\varphi \propto [y(1-y)]^{(N-3)/2}\mathrm{d}y$ — so we sample

$$y \sim \text{Beta}\Big(\tfrac{N-1}{2},\,\tfrac{N-1}{2}\Big) \text{ truncated to } y \in [(1+\varphi_c)/2,\,1], \qquad \varphi_{1\mu} = 2y - 1, \tag{12}$$

which standard libraries provide directly. Each live pattern is constructed from $\varphi_{1\mu}$ and a uniform direction u_μ on the $(N-2)$-sphere perpendicular to $\boldsymbol{\xi}^1$: $\boldsymbol{\xi}^\mu = \sqrt{N}\,(\varphi_{1\mu},\,\sqrt{1-\varphi_{1\mu}^2}\,u_\mu)$ in the basis where $\boldsymbol{\xi}^1 = (\sqrt{N}, 0, \ldots, 0)$. The live set is fixed once per disorder.

Metropolis runs on the full spin $\boldsymbol{x} \in \sqrt{N}\cdot S^{N-1}$: Gaussian proposal $\boldsymbol{x}' = \boldsymbol{x} + \sigma\boldsymbol{\eta}$ projected back to $|\boldsymbol{x}'| = \sqrt{N}$; energy $H(\boldsymbol{x}) = -\beta_{\text{net}}^{-1}\ln\!\big[\mathrm{e}^{\beta_{\text{net}}\,\boldsymbol{\xi}^1\boldsymbol{x}} + \sum_{\mu=2}^{K_{\text{cusp}}+1}\mathrm{e}^{\beta_{\text{net}}\,\boldsymbol{\xi}^\mu\boldsymbol{x}} + M\,\mathrm{e}^{Ng_{\max}(\beta_{\text{net}})}\big]$. Cost per step $O(K_{\text{cusp}}\cdot N)$.

Crucially the bulk being a constant in $\boldsymbol{x}$ does not make the free energy constant in φ_1: the chain still feels the sphere geometry, and below the cusp (10) gives $F(\varphi_1)/N = -\varphi_c - (T/2)\ln(1-\varphi_1^2)$, which is minimised at $\varphi_1 = 0$. Once the chain crosses the cusp it slides toward $\varphi_1 = 0$ under this entropic attraction; that slide is what we use to detect basin escape (Sec. 4).

At $N = 100$ the spherical density falls fast enough that the expected K_{cusp} at the cusp is below 1 for every $\alpha \in [0.5, 0.62]$ we measure. The algorithm therefore keeps only $\boldsymbol{\xi}^1$ explicit and the bulk constant carries the rest — the chain samples the leading-order partition function (10) directly, with no $1/N$ subleading corrections.

Why naive “smart” fails. A faster variant — “smart” — refreshes its K kept patterns every N_{REFRESH} steps from a cone around the current chain. This breaks the quenched-disorder assumption. As the chain wanders, fresh patterns are minted at every refresh; the total set of distinct patterns seen along one trajectory exceeds M, and the chain effectively faces a larger

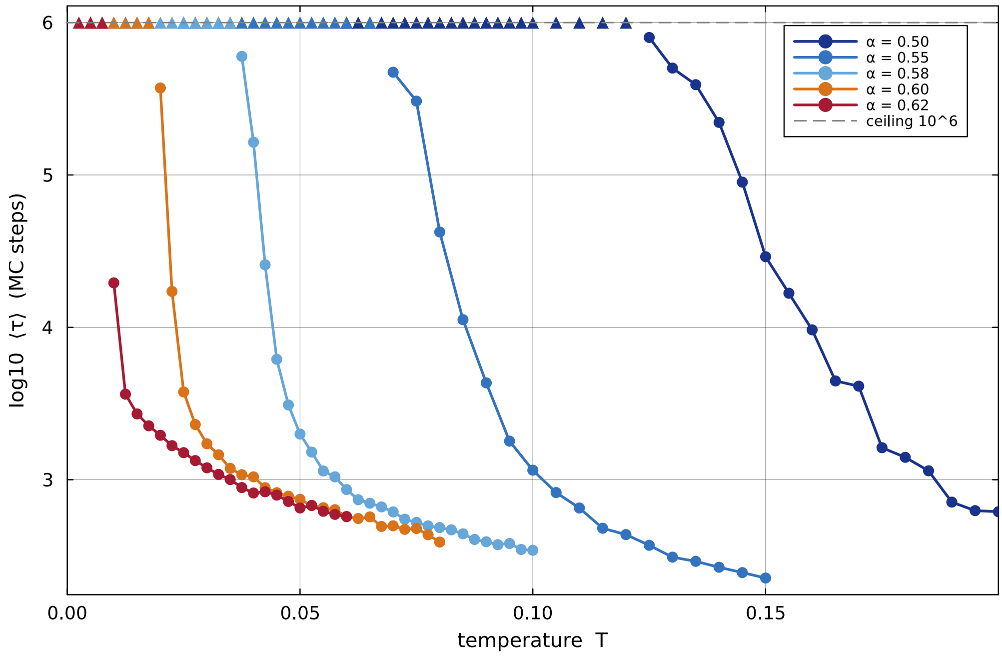

Figure 2: Mean basin escape time $\langle\tau\rangle$ vs T at $N = 100$ from N-dim hybrid MC, 32 disorders, $\beta_{\rm net} = 1$. Five α values: $0.50, 0.55, 0.58, 0.60, 0.62$. Filled circles: mean over disorders that escaped within 10^6 MC steps. Upward triangles at $\log_{10}\tau = 6$: cells where no disorder escaped (lower bound only). The escape curve slides toward $T = 0$ as $\alpha \to 1 - g_{\max} \approx 0.6226$.

pattern set than the model defines. Near the retrieval basin the extra competitors destroy the basin. Both truncated and hybrid avoid the problem: the retained set is fixed once per disorder, no refresh.

4 Results

We measure two observables at $N = 100$: the mean first-passage escape time τ from the retrieval basin, and the steady-state $\langle\varphi_1\rangle - \phi_{\rm eq}(T)$ residual.

Escape time. Figure 2 reports $\log_{10}\langle\tau\rangle$ vs T at $N = 100$ for five α values from the N-dim hybrid MC, averaged over 32 disorders, with a budget cap at 10^6 MC steps. Each curve is a basin escape profile: at low T the basin is stable for the full budget (upward triangles at the ceiling); above a critical $T_{\rm esc}(\alpha)$ the escape time drops by several orders of magnitude over a T window of width ~ 0.01. $T_{\rm esc}$ moves to lower T as α rises; extrapolating to $\tau \to \infty$ at $T = 0$ gives a critical load near $\alpha = 1 - g_{\max} \approx 0.6226$.

Residual heatmap. Figure 3 shows $\langle\varphi_1\rangle - \phi_{\rm eq}(T)$ over the (α, T) plane. Background: truncated MC at varying N over $\alpha \in [0.20, 0.70]$. Grey rectangle: N-dim hybrid MC at $N = 100$ on a fine T grid in $\alpha \in [0.50, 0.62]$, $T \in [0, 0.10]$. The basin (white) is bounded by a sharp transition (dark red) that follows the saddle line $\alpha_c^{\rm sd}(T)$ and pinches at $\alpha \approx 0.6226$, $T = 0$.

5 Discussion

The Ramsauer threshold $\alpha_c = 1/2$ is a Gaussian artefact. The Gaussian density misses the interior saddle of the partition-sum integrand (4) that the exact spherical density places at $\varphi^*(\beta_{\rm net})$. Including it gives $\alpha_c(\beta_{\rm net}, 0) = \beta_{\rm net} - g_{\max}(\beta_{\rm net})$; at the value $\beta_{\rm net} = 1$ used in our MC, $\alpha_c \approx$

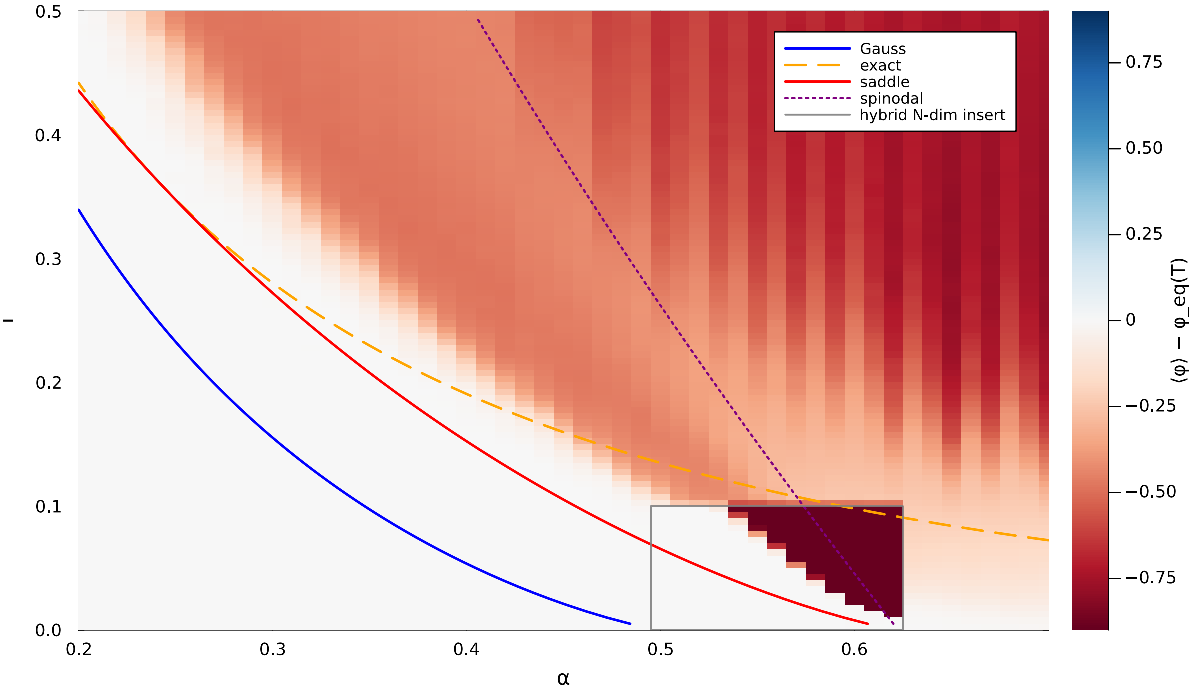

Figure 3: $\langle\varphi_1\rangle - \phi_{\rm eq}(T)$ from Monte Carlo over the (α, T) plane. Curves: $\alpha_c^{\rm G}(T)$ (blue solid, Eq. 8, Gaussian density + support edge); $\alpha_c^{\rm bd}(T)$ (orange dashed, Eq. 9, exact density + support edge); $\alpha_c^{\rm sd}(T)$ (red solid, Eq. 7, exact density + interior saddle, this work); $\alpha_c^{\rm sp}(T)$ (purple dotted, Eq. 11, single-pattern spinodal). Background: truncated MC at varying N over the full α range, linearly interpolated to the fine T grid. Grey rectangle: N-dim hybrid MC at $N = 100$ on the fine T grid. The basin pinches at $\alpha \approx 0.6226$ — Ramsauer's $\alpha_c^{\rm G}(0) = 1/2$ undershoots by 0.12 in α, a factor 1.25 in capacity.

0.6226, a factor 1.25 larger than Ramsauer's 1/2. The shift is per spin, not asymptotic: it appears at leading order in N and is visible already at $N = 100$. A second signature of the artefact is that Ramsauer's 1/2 is $\beta_{\rm net}$-independent, while the exact-density (7) boundary is not.

Open items. (i) Hot init. All present runs start the chain at $\boldsymbol{\xi}^1$ (cold). The basin boundary they trace is the spinodal, not the thermodynamic crossover; hot init at the same grid would delimit the metastability band. (ii) Cusp rounding at finite N. The leading-order $\max(\cdot,\cdot)$ in (10) is rounded on a scale $1/N$ in φ_1. Near the endpoint $(0.6226, 0)$ the basin offset $\phi_{\rm eq}(T) - (\alpha + g_{\max})$ falls inside the rounding window, and the leading-order spinodal slope needs an NLO correction. (iii) Heatmap consolidation. Figure 3 stitches three data sources. A clean version would use a single MC variant — truncated with a hybrid insert near the cusp — across the entire (α, T) plane.

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